SERRE'S CRITERION FOR AFFINENESS IN THE SETTING OF ALGEBRAIC SPACES, BY A PRESENTATION OF THE STRUCTURE SHEAF OF AN AFFINE ETALE COVER

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The vanishing situation is separated: the fibre product of an affine etale cover is a closed subscheme of the product of the base with itself. Same chapter spaces-cohomology.tex, label lemma-vanishing-separated (invoked at proof line 3173). It makes $R = U \times_X U$ a closed subscheme of $\Spec(B \otimes_A B)$, hence affine, which is what allows the presentation of the algebra C.
- The base change identification of the pullback of the pushforward of the structure sheaf. Same chapter spaces-cohomology.tex, label lemma-flat-base-change-cohomology (invoked at proof line 3191). It identifies $\varphi^*\varphi_*\mathcal{O}_U
with $t_*\mathcal{O}_R$, so that surjectivity of the map on the cover becomes the generation condition $s(g_i)$ over t.
- The injectivity consequence of the vanishing situation, and representability of a separated locally quasi-finite morphism. Same chapter spaces-cohomology.tex, label lemma-vanishing-injective (invoked at proof line 3201) and Morphisms of Spaces, labels spaces-morphisms-lemma-locally-quasi-finite and spaces-morphisms-lemma-locally-quasi-finite-separated-representable (invoked at proof lines 3203 and 3206). Together with separatedness and finite type they make $X \to \Spec(A)$ representable, so that X is a scheme.
- The affine criterion for a quasi-compact separated scheme on which H^1 of every quasi-coherent module vanishes. Cohomology of Schemes, label coherent-lemma-quasi-compact-h1-zero-covering (invoked at proof line

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3208). It finishes the argument by showing that the scheme X is affine, hence equal to the base.

Source reference. Original lemma 01XF: sources/stacks-chow/tags.txt, source lines 2516–2516. Navigation pointer for \ref{coherent-lemma-quasi-compact-h1-zero-covering} used inside the counted span (line 3209); the argument that consumes it is retained in the span.

Source reference. Original lemma 073K: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1922–1922. Navigation pointer for \ref{lemma-flat-base-change-cohomology} used inside the counted span (line 3191); the argument that consumes it is retained in the span.

Source reference. Original lemma 07V4: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3097–3097. Navigation pointer for \ref{lemma-vanishing-injective} used inside the counted span (line 3201); the argument that consumes it is retained in the span.

Source reference. Original lemma 07V5: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3132–3132. Navigation pointer for \ref{lemma-vanishing-separated} used inside the counted span (line 3173); the argument that consumes it is retained in the span.

Source reference. Original reference 07UY: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2991–2991. Navigation pointer for \ref{situation-vanishing} used inside the counted span (line 3167); the argument that consumes it is retained in the span.

Source reference. Original lemma 06RW: sources/stacks-chow/tags.txt, source lines 8655–8655. Navigation pointer for \ref{spaces-morphisms-lemma-locally-quasi-finite} used inside the counted span (line 3203); the argument that consumes it is retained in the span.

Source reference. Original lemma 0418: sources/stacks-chow/tags.txt, source lines 5283–5283. Navigation pointer for \ref{spaces-morphisms-lemma-locally-quasi-finite-separated-representable} used inside the counted span (line 3206); the argument that consumes it is retained in the span.

Proposition 0.1. *A quasi-compact and quasi-separated algebraic space is affine if and only if all higher cohomology groups of quasi-coherent sheaves vanish. More precisely, any algebraic space as in Situation 07UY is an affine scheme.*

Proof. Choose an affine scheme $U = \mathrm{Spec}(B)$ and a surjective étale morphism $\varphi : U \to X$. Set $R = U \times_X U$. As p is separated (Lemma 07V5) we see that R is a closed subscheme of $U \times_{\mathrm{Spec}(A)} U = \mathrm{Spec}(B \otimes_A B)$. Hence $R = \mathrm{Spec}(C)$ is affine too and the ring map

$$B \otimes_A B \longrightarrow C$$

is surjective. Let us denote the two maps $s, t : B \to C$ as usual. Pick $g_1, \ldots, g_m \in B$ such that $s(g_1), \ldots, s(g_m)$ generate C over $t : B \to C$ (which is possible as $t : B \to C$ is of finite presentation and the displayed map is surjective). Then $g_1, \ldots, g_m$ give global sections of $\varphi_*\mathcal{O}_U$ and the map

$$\mathcal{O}_X[z_1, \ldots, z_n] \longrightarrow \varphi_*\mathcal{O}_U, \quad z_j \longmapsto g_j$$

is surjective: you can check this by restricting to U. Namely, $\varphi^*\varphi_*\mathcal{O}_U = t_*\mathcal{O}_R$ (by Lemma 073K) hence you get exactly the condition that $s(g_i)$ generate C over $t : B \to C$. By the vanishing of H^1 of the kernel we see that

$$\Gamma(X, \mathcal{O}_X[x_1, \ldots, x_n]) = A[x_1, \ldots, x_n] \longrightarrow \Gamma(X, \varphi_*\mathcal{O}_U) = \Gamma(U, \mathcal{O}_U) = B$$

is surjective. Thus we conclude that B is a finite type A-algebra. Hence $X \to \mathrm{Spec}(A)$ is of finite type and separated. By Lemma 07V4 and Morphisms of Spaces, Lemma 06RW it is also locally quasi-finite. Hence $X \to \mathrm{Spec}(A)$ is representable by Morphisms of Spaces, Lemma 0418 and X is a scheme. Finally X is affine, hence equal to $\mathrm{Spec}(A)$, by an application of Cohomology of Schemes, Lemma 01XF. $\square$